Bright Tuition class

Subject : Math, Phy, Che
Standard : 11
Total Mark : 300

Jee paper std 11 8 th February

Paper Set : 1
Date : 27-02-2026
Time : 0H:0M

Mathematics - Section A (MCQ)

(1) $\cos\frac{\pi}{5}\cos\frac{2\pi}{5}\cos\frac{4\pi}{5}\cos\frac{8\pi}{5} =$
(A) $1/16$ (B) 0
(C) $-1/8$ (D) $-1/16$

(2) Let $a_1, a_2, a_3, \ldots$ be an $A.P.$ If $\frac{a_1+a_2+\ldots+a_{10}}{a_1+a_2+\ldots+a_p} = \frac{100}{p^2}, p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to :
(A) $\frac{19}{21}$ (B) $\frac{100}{121}$
(C) $\frac{21}{19}$ (D) $\frac{121}{100}$

(3) If $\sin A = \frac{1}{\sqrt{10}}$ and $\sin B = \frac{1}{\sqrt{5}}$, where A and B are positive acute angles, then $A + B =$
(A) π (B) $\pi/2$
(C) $\pi/3$ (D) $\pi/4$

(4) If $\left(\frac{1}{\alpha+1} + \frac{1}{\alpha+2} + \ldots + \frac{1}{\alpha+1012}\right) - \left(\frac{1}{2\cdot1} + \frac{1}{4\cdot3} + \frac{1}{6\cdot5} + \ldots + \frac{1}{2024\cdot2023}\right) = \frac{1}{2024}$, then α is equal to-
(A) 1367 (B) 1058
(C) 1056 (D) 1011

(5) Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}} + \beta x\right)^4, (1-3\beta x)^2$ and $\left(1 - \frac{\beta}{2}x\right)^6, \beta > 0$, respectively form the first three terms of an $A.P.$ If d is the common difference of this $A.P.$, then $50 - \frac{2d}{\beta^2}$ is equal to.
(A) 57 (B) 56
(C) 55 (D) 54

(6) The equation $(a+b)^2 = 4ab\sin^2\theta$ is possible only when
(A) $2a = b$ (B) $a = b$
(C) $a = 2b$ (D) None of these

(7) The term independent of x in the expansion of $(1+x)^n\left(1+\frac{1}{x}\right)^n$ is
(A) $C_0^2 + 2C_1^2 + \ldots. + (n+1)C_n^2$
(B) $(C_0 + C_1 + \ldots. + C_n)^2$
(C) $C_0^2 + C_1^2 + \ldots.. + C_n^2$
(D) None of these

(8) If number of terms in the expansion of $(x - 2y + 3z)^n$ are 45, then $n =$
(A) 7 (B) 8
(C) 9 (D) None of these

(9) $x^5 + 10x^4a + 40x^3a^2 + 80x^2a^3 + 80xa^4 + 32a^5 =$
(A) $(x+a)^5$ (B) $(3x+a)^5$
(C) $(x+2a)^5$ (D) $(x+2a)^3$

(10) Let the coefficient of x^r in the expansion of $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \ldots\ldots + (x+2)^{n-1}$ be α_r. If $\sum_{r=0}^{n}\alpha_r = \beta^n - \gamma^n, \beta, \gamma \in N$, then the value of $\beta^2 + \gamma^2$ equals.....
(A) 23 (B) 24
(C) 20 (D) 25

(11) If the arithmetic mean of two numbers be A and geometric mean be G, then the numbers will be
(A) $A \pm (A^2 - G^2)$ (B) $\sqrt{A} \pm \sqrt{A^2 - G^2}$
(C) $A \pm \sqrt{(A+G)(A-G)}$ (D) $\frac{A \pm \sqrt{(A+G)(A-G)}}{2}$

(12) $n^n\left(\frac{n+1}{2}\right)^{2n}$ is
(A) Less than $\left(\frac{n+1}{2}\right)^3$ (B) Greater than $\left(\frac{n+1}{2}\right)^3$
(C) Greater than $(n!)^3$ (D) (b) and (c) both

(13) Let $a, b, c > 1, a^3, b^3$ and c^3 be in $A.P.$, and $\log_a b$, $\log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an $A.P.$, whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is -444, then abc is equal to
(A) 343 (B) 216
(C) $\frac{343}{8}$ (D) $\frac{125}{8}$

(14) Fifth term of a $G.P.$ is 2, then the product of its 9 terms is
(A) 256 (B) 512
(C) 1024 (D) None of these

(15) If $p = \frac{2\sin\theta}{1+\cos\theta+\sin\theta}$, and $q = \frac{\cos\theta}{1+\sin\theta}$, then
(A) $pq = 1$ (B) $\frac{q}{p} = 1$
(C) $q - p = 1$ (D) $q + p = 1$

(16) The value of $36(4\cos^2 9^\circ - 1)(4\cos^2 27^\circ - 1)(4\cos^2 81^\circ - 1)(4\cos^2 243^\circ - 1)$ is
(A) 54 (B) 18
(C) 27 (D) 36

(17) The sum of all two digit numbers which, when divided by 4, yield unity as a remainder is
(A) 1190 (B) 1197
(C) 1210 (D) None of these

(18) If $T_0, T_1, T_2, \ldots. T_n$ represent the terms in the expansion of $(x+a)^n$, then $(T_0 - T_2 + T_4 - \ldots.)^2 + (T_1 - T_3 + T_5 - \ldots.)^2 =$
(A) $(x^2 + a^2)$ (B) $(x^2 + a^2)^n$
(C) $(x^2 + a^2)^{1/n}$ (D) $(x^2 + a^2)^{-1/n}$

(19) $2\cos^2\theta - 2\sin^2\theta = 1$, then $\theta =$ .....^{o}
(A) 15 (B) 30
(C) 45 (D) 60

(20) $2 + 4 + 7 + 11 + 16 + \ldots\ldots$ to n terms =
(A) $\frac{1}{6}(n^2 + 3n + 8)$ (B) $\frac{n}{6}(n^2 + 3n + 8)$
(C) $\frac{1}{6}(n^2 - 3n + 8)$ (D) $\frac{n}{6}(n^2 - 3n + 8)$

Mathematics - Section B (NUMERIC)

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(21) The coefficient of x in the expansion of $[\sqrt{1+x^2}-x]^{-1}$ in ascending powers of x, when $|x|<1$, is

(22) Let $75\ldots57$ denote the $(r+2)$ digit number where the first and the last digits are 7 and the remaining r digits are 5. Consider the sum $S=77+757+7557+\ldots+75\ldots.57$. If $S=\frac{75^{98}\ldots57+m}{n}$, where m and n are natural numbers less than 3000, then the value of $m+n$ is

(23) $\tan 9^\circ-\tan 27^\circ-\tan 63^\circ+\tan 81^\circ=$

(24) Consider a sequence whose sum of first n -terms is given by $S_n=4n^2+6n, n\in N$, then T_{15} of this sequence is -

(25) If $S=\frac{7}{5}+\frac{9}{5^2}+\frac{13}{5^3}+\frac{19}{5^4}+\ldots.$, then 160 S is equal to..... .

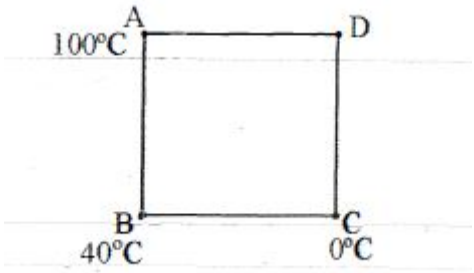
Physics - Section A (MCQ)

(26) A pendulum clock loses 12 sec a day if the temperature is $40^\circ C$ and gains 4 sec a day if the temperature is $20^\circ C$. The temperature at $\ldots^\circ C$ clock will show correct time.

(A) 25 (B) 60

(C) 30 (D) 55

(27) Four conducting rods are joined to make a square. All rods are identical and ends A, B and C are maintained at given temperatures. choose $INCORRECT$ statement for given arrangement in steady state. (value of $\frac{KA}{L}$ is $1\frac{J}{S^\circ C}$, symbols , have their usual meaning)



(A) Heat current in AB is equal to 1.5 times of heat current in BC

(B) Temperature of end D is $50\,^oC$

(C) Heat current in AB is equal to heat current in BC

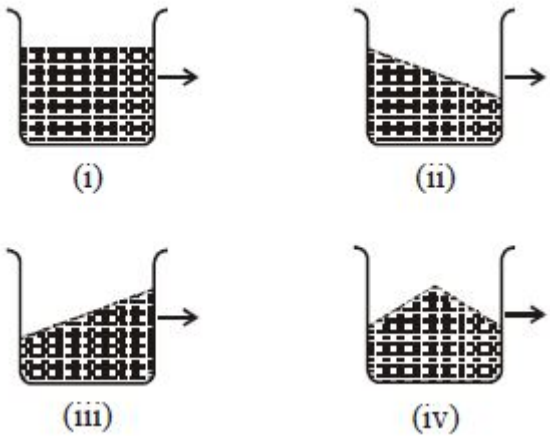
(D) Heat current withdrawn at end B is $20\ J/S$

(28) Two mercury drops (each of radius r) merge to form a bigger drop. The surface energy of the bigger drop, if T is the surface tension is

(A) $2^{5/3}\pi r^2T$ (B) $4\pi r^2T$

(C) $2\pi r^2T$ (D) $2^{8/3}\pi r^2T$

(29) A vessel containing water is moving with a constant speed towards right along a straight horizontal path. Which of the following diagrams represents the surface of liquid?



(A) (i) (B) (ii)

(C) (iii) (D) (iv)

(30) A soap film of surface tension $3\times10^{-2}\,Nm^{-1}$ formed in rectangular frame, can support a straw. The length of the film is $10\,cm$. Mass of the straw the film can support is ..... gm

(A) 0.06 (B) 0.6

(C) 6 (D) 60

(31) In a test experiment on a model aeroplane in a wind tunnel, the flow speeds on the upper and lower surfaces of the wing are $70\ ms^{-1}$ and $63\ ms^{-1}$ respectively. What is the lift on the wing if its area is $2.5\ m^2$? Take the density of atr to be $1.3\ kgm^{-3}$.

(A) $1.51\times10^3\ N$ (B) $3.64\times10^4\ N$

(C) $2.67\times10^5\ N$ (D) $8.60\times10^2\ N$

(32) A metallic ball and highly stretched spring are made of the same material and have the same mass. They are heated so that they melt, the latent heat required

(A) Are the same for both

(B) Is greater for the ball

(C) Is greater for the spring

(D) For the two may or may not be the same depending upon the metal

(33) When heat energy of $1500\ Joules$, is supplied to a gas at constant pressure $2.1\times10^5\ N/m^2$, there was an increase in its volume equal to $2.5\times10^{-3}\ m^3$. The increase in internal energy of the gas in Joules is ..... J

(A) 450 (B) 525

(C) 975 (D) 2025

(34) What will be the work done in increasing the radius of soap bubble from $\frac{r}{2}$ to $2r$, if surface tension of soap solution is T ?

(A) $3\pi r^2T$ (B) $15\pi r^2T$

(C) $9\pi r^2T$ (D) $12\pi r^2T$

(35) The ratio of radiant energies radiated per unit surface area by two bodies is $16:1$, the temperature of hotter body is $1000K$, then the temperature of colder body will be ..... K

(A) 250 (B) 500

(C) 1000 (D) 62.5

(36) A cube of wood supporting $200\,gm$ mass just floats in water. When the mass is removed, the cube rises by $2\,cm$. ..... cm is the side of cube .

(A) 5 (B) 10

(C) 15 (D) 20

(37) An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.

A. The work done by gas during the process is zero.

B. The heat added to gas is different from change in its internal energy.

C. The volume of the gas is increased.

D. The internal energy of the gas is increased.

E. The process is isochoric (constant volume process)

Choose the correct answer from the options given below : $-$

(A) A, B, C, D Only (B) A, D, E Only

(C) E Only (D) A, C Only

(38) A black body at $200K$ is found to exit maximum energy at a wavelength of $14\mu m$. When its temperature is raised to $1000K$, the wavelength at which maximum energy is

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emitted is

(A) $14\mu m$ (B) $70\mu F$

(C) $2.8\mu m$ (D) $2.8mm$

(39) In rainy season, on a clear night the black seat of a bicycle becomes wet because

(A) It absorbs water vapour

(B) Black seat is good absorber of heat

(C) Black seat is good radiator of heat energy

(D) None of the above

(40) A liquid drop of diameter D breaks into 27 tiny drops. The change in energy is

(A) $2\pi TD^2$ (B) $4\pi TD^2$

(C) πTD^2 (D) None of these

(41) For an isothermal expansion of a perfect gas, the value of $\frac{\Delta P}{P}$ is equal

(A) $-\gamma^{1/2}\frac{\Delta V}{V}$ (B) $-\frac{\Delta V}{V}$

(C) $-\gamma\frac{\Delta V}{V}$ (D) $-\gamma^2\frac{\Delta V}{V}$

(42) In supplying 400 calories of heat to a system, the work done will be

(A) 400 joules (B) 1672 joules

(C) 1672 watts (D) 1672 ergs

(43) A container of height $10\,cm$ is filled with water. There is a hole at bottom. Find the pressure difference between points at top and bottom.

(A) $1000\,Pa$ (B) zero

(C) $1\,Pa$ (D) $100\,Pa$

(44) Assertion :- A brass disc is fast fitted in a hole in a steel plate. The system must be cooled to loosen the disc from the hole.
Reason :- The coefficient of linear expansion of brass, is greater than the coefficient of linear expansion of steel.

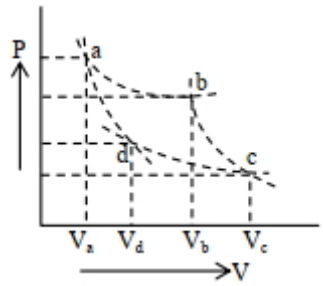
(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

(45) Two different adiabatic paths for the same gas intersect two isothermal curves as shown in $P-V$ diagram. The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:



(A) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^{-1}$ (B) $\frac{V_a}{V_d}\neq\frac{V_b}{V_c}$

(C) $\frac{V_a}{V_d}=\frac{V_b}{V_c}$ (D) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^2$

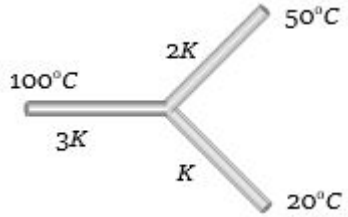
Physics - Section B (NUMERIC)

(46) A ball whose density is $0.4\times10^3\,kg/m^3$ falls into water from a height of $9\,cm$. To what depth does the ball sink ? ..... cm

(47) A big drop is formed by coalescing 1000 small identical drops of water. If E_1 be the total surface energy of 1000 small drops of water and E_2 be the surface energy of single big drop of water, the E_1 : E_2 is x : 1 where x =_____.

(48) If heat is added at constant volume, $6300\ J$ of heat are required to raise the temperature of an ideal gas by $150\ K$. If instead, heat is added at constant pressure, 8800 joules are required for the same temperature change. When the temperature of the gas changes by $300\ K$, the internal energy of the gas changes by ..... J

(49) Three rods of the same dimension have thermal conductivities $3K$, $2K$ and K . They are arranged as shown in fig. Given below, with their ends at $100^oC, 50^oC$ and 20^oC. The temperature of their junction is ..... ^oC

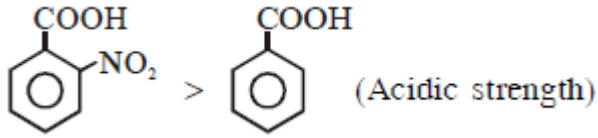


(50) A faulty thermometer has its fixed points marked 5° and 95°. This thermometer reads the temperature of a body as 59°. Then the correct temperature of the body on celsius scale is ..... ^{o}

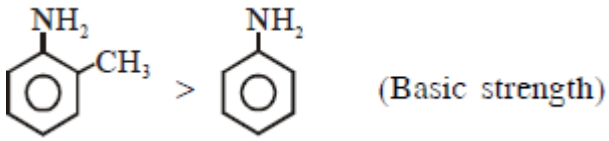
Chemistry - Section A (MCQ)

(51) Incorrect order is

(A)

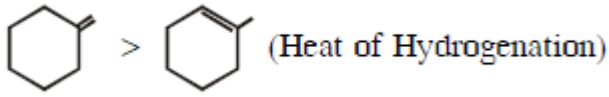


(B)

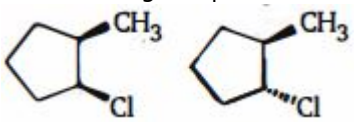


(C) $CH_3-\underset{\underset{CH_3}{|}}{\overset{CH_3}{|}}{C}-O^{\ominus} > CH_3-\overset{\ominus}{C}H_2$ (Stability)

(D)



(52) The following compounds are identical with respect to



(A) molecular composition (B) boiling point

(C) melting point (D) $IUPAC$ name

(53) In the given molecule $(NC)_3CCCCHCHBr$ number of σ and $\pi-$ bonds respectively are

(A) 13 and 9 (B) 12 and 8

(C) 7 and 3 (D) 10 and 5

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(54) In the sulphonation, acetylation and formylation of benzene the group of effective electrophiles would be :

(A) $SO_3^+, CH_3 \equiv \overset{+}{O}, H\overset{+}{C}O$

(B) $SO_3, CH_3 - C \equiv \overset{+}{O}, H\overset{+}{C}O$

(C) $SO_3, CH_3CHO, CO + HCl$

(D) HSO_3, CH_3CO, HCO

(55) Number of σ and π bonds present in ethylene molecule is respectively :

(A) 3 and 1 (B) 5 and 2

(C) 4 and 1 (D) 5 and 1

(56) Which of the following compounds will show geometrical isomerism

(A) $2-$ Butene (B) Propene

(C) Ethylene (D) $2-$ Methyl $-2-$ butene

(57) Geometrical isomerism is shown by

(A) $2-$butene (B) $2-$butyne

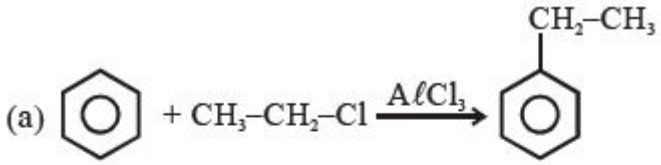
(C) $2-$butanol (D) Butanal

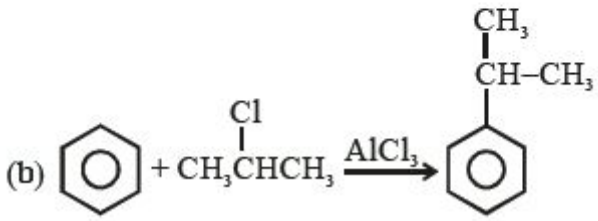
(58) The halogen which is most reactive in the halogenation of alkenes under sunlight is

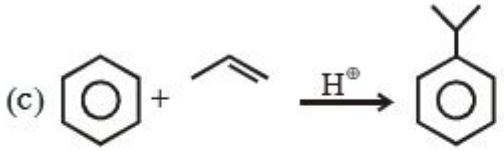
(A) Chlorine (B) Bromine

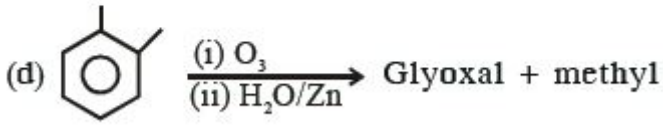
(C) Iodine (D) All equal

(59) Correctly matched products are

(a) 

(b) 

(c) 

(d) 

glyoxal + Dimethyl glyoxal

(A) b, c (B) b, c, d

(C) a, b, c, d (D) a, b, d

(60) The $IUPAC$ name of $\underset{COOH}{\underset{|}{CH_2}} - CH_2 - \underset{COOH}{\underset{|}{CH_2}}$ is

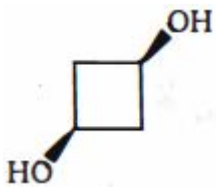
(A) Pentane $-1, 5-$ dioic acid

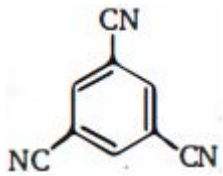
(B) Pentane $-1, 5-$ dicarboxylic acid

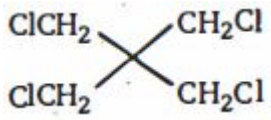
(C) Propane $-1, 3-$ dioic acid

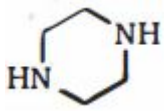
(D) none of these

(61) Which one of the following compounds has non zero dipole moment?

(A) 

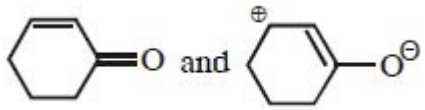
(B) 

(C) 

(D) 

(62) Which of the following pairs does not represent resonating structures ?

(A) $CH_3 - \overset{O}{\overset{||}{C}} - CH_3$ and $CH_3 - \overset{OH}{\overset{|}{C}} = CH_2$

(B) 

(C) $CH_3 - \overset{\overset{\oplus}{O}-H}{\overset{||}{C}} - CH_3$ and $CH_3 - \underset{\oplus}{\overset{OH}{\overset{|}{C}}} - CH_3$

(D) $CH_2 = C = O$ and $\overset{\ominus}{C}H_2 - C \equiv \overset{\oplus}{O}$

(63) The structure of $4-$ methyl pentene -2 is

(A) $(CH_3)_2CH - CH_2CH = CH_2$

(B) $(CH_3)_2CH - CH = CH - CH_3$

(C) $(CH_3)_2CH - CH_2CH = CH - CH_3$

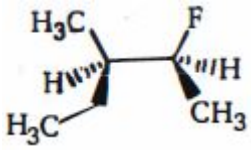
(D) $(CH_3)_2C = CHCH_2CH_3$

(64) Which is an electrophile

(A) $AlCl_3$ (B) CN^-

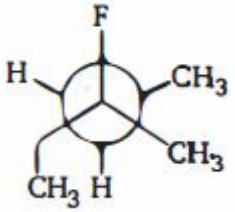
(C) NH_3 (D) CH_3OH

(65) Which is the most stable conformer along the $2, 3\, C - C$ bond axis of the compound ?

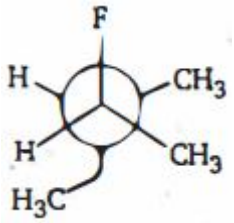


tion class

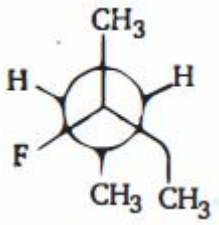
(A)



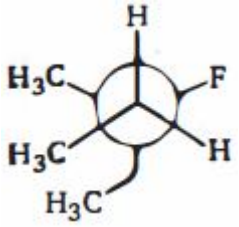
(B)



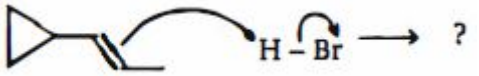
(C)



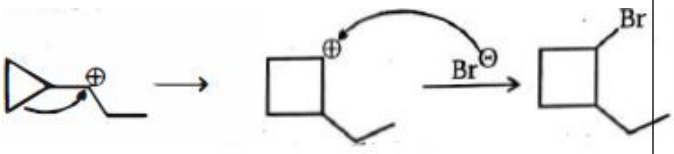
(D)



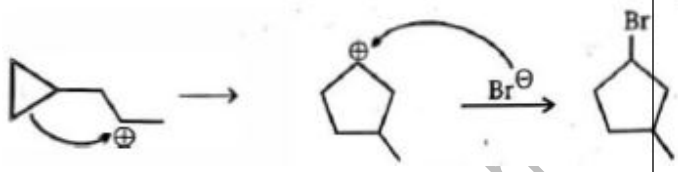
(66) Taking into account the stability of various carbocations and, as well as the rules governing mechanisms of carbocation rearrangements, which reaction is most likely to occur during the given reaction ?



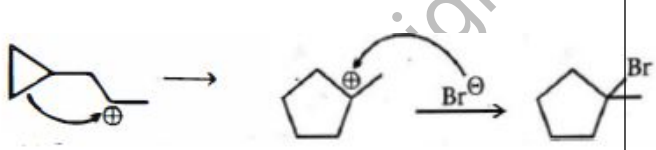
(A)



(B)



(C)



(D) None

(67) The $IUPAC$ name of $\begin{array}{ccccc} CH_3 & - CH & - C & - C & - NH \\ & | & \| & \| & | \\ & CH_3 & O & O & Br \end{array}$ is

(A) $(N-$ Bromo $)-2-$ keto $-3-$ methylbutanamide

(B) $(N-$ Bromo $)-2-$ keto $-4-$ methylbutanamide

(C) $(N-$ Bromo $)-1,2-$ diketo $-3-$ methylbutanamine carboxamide

(D) $(N-$ Bromo$)-1-$ keto $-2-$ methylpropane

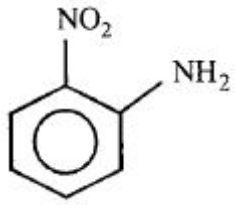
(68) Minimum number of carbon atoms required for compound to show positional isomerism ?

(A) 4 (B) 3

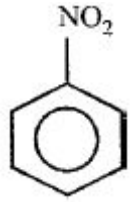
(C) 2 (D) 1

(69) Which compound exhibits maximum dipole moment among the following ?

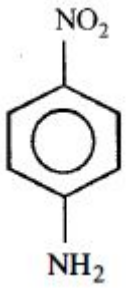
(A)



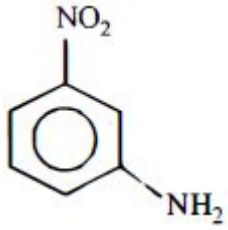
(B)



(C)



(D)



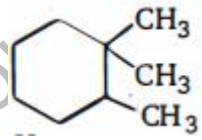
(70) Amongst the following, the compound that can be most readily sulphonated is

(A) Benzene (B) Nitrobenzene

(C) Toluene (D) Chlorobenzene

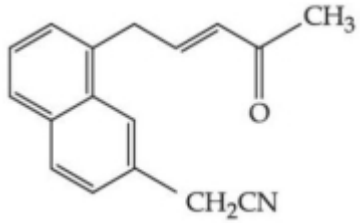
Chemistry - Section B (NUMERIC)

(71) How many geometrical isomers are possible for the above compound ?



(72) How many secondary carbon atoms does methyl cyclopropane have ?

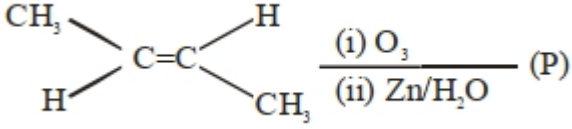
(73) Number of electrophilic centre in the given compound is ..... .



(74) $3-Methylhex-2-$ene on reaction with HBr in presence of peroxide forms an addition product (A). The number of possible stereoisomers for ' A ' is_____.

(75) $Image$

Consider the given reaction. The total number of oxygen atoms present per molecule of the product (P) is_____



Bright Tuition class

Subject : Math, Phy, Che
Standard : 11
Total Mark : 300

Jee paper std 11 8 th February

(Answer Key)

Paper Set : 1
Date : 27-02-2026
Time : 0H:0M

Mathematics - Section A (MCQ)

1 - D	2 - C	3 - D	4 - D	5 - A	6 - B	7 - C	8 - B	9 - C	10 - D
11 - C	12 - D	13 - B	14 - B	15 - D	16 - D	17 - C	18 - B	19 - B	20 - B

Mathematics - Section B (NUMERIC)

21 - 1	22 - 1219	23 - 4	24 - 122	25 - 305

Physics - Section A (MCQ)

26 - A	27 - C	28 - D	29 - A	30 - B	31 - A	32 - A	33 - C	34 - B	35 - B
36 - B	37 - B	38 - C	39 - C	40 - A	41 - B	42 - B	43 - A	44 - A	45 - C

Physics - Section B (NUMERIC)

46 - 6	47 - 10	48 - 12600	49 - 70	50 - 60

Chemistry - Section A (MCQ)

51 - B	52 - A	53 - A	54 - B	55 - D	56 - A	57 - A	58 - A	59 - C	60 - A
61 - A	62 - A	63 - B	64 - A	65 - B	66 - D	67 - A	68 - C	69 - C	70 - C

Chemistry - Section B (NUMERIC)

71 - 0	72 - 2	73 - 3	74 - 4	75 - 1

Bright Tuition class

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Subject : Math, Phy, Che
Standard : 11
Total Mark : 300

Jee paper std 11 8 th February

(Solutions)

Paper Set : 1
Date : 27-02-2026
Time : 0H:0M

Mathematics - Section A (MCQ)

(1) $\cos\frac{\pi}{5}\cos\frac{2\pi}{5}\cos\frac{4\pi}{5}\cos\frac{8\pi}{5} =$

(A) $1/16$ (B) 0

(C) $-1/8$ **(D)** $-1/16$

Solution:(Correct Answer:D)

(d) $\cos\frac{\pi}{5}\cos\frac{2\pi}{5}\cos\frac{4\pi}{5}\cos\frac{8\pi}{5}$

$= \frac{\sin\frac{2^4\pi}{5}}{2^4\sin\frac{\pi}{5}} = \frac{\sin\frac{16\pi}{5}}{16\sin\frac{\pi}{5}}$

$= \frac{\sin\left(3\pi+\frac{\pi}{5}\right)}{16\sin\frac{\pi}{5}}$

$= \frac{-\sin\frac{\pi}{5}}{16\sin\frac{\pi}{5}} = -\frac{1}{16}.$

(2) Let $a_1, a_2, a_3, \ldots$ be an $A.P.$ If $\frac{a_1+a_2+\ldots+a_{10}}{a_1+a_2+\ldots+a_p} = \frac{100}{p^2}, p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to :

(A) $\frac{19}{21}$ (B) $\frac{100}{121}$

(C) $\frac{21}{19}$ (D) $\frac{121}{100}$

Solution:(Correct Answer:C)

$\frac{\frac{10}{2}(2a_1+9\,d)}{\frac{p}{2}(2a_1+(p-1)d)} = \frac{100}{p^2}$

$(2a_1+9\,d)\,p = 10\,(2a_1+(p-1)d)$

$9dp = 20a_1 - 2pa_1 + 10\,d(p-1)$

$9p = (20-2p)\frac{a_1}{d} + 10(p-1)$

$\frac{a_1}{d} = \frac{(10-p)}{2(10-p)} = \frac{1}{2}$

$\therefore \frac{a_{11}}{a_{10}} = \frac{a_1+10\,d}{a_1+9\,d} = \frac{\frac{1}{2}+10}{\frac{1}{2}+9} = \frac{21}{19}$

(3) If $\sin A = \frac{1}{\sqrt{10}}$ and $\sin B = \frac{1}{\sqrt{5}}$, where A and B are positive acute angles, then $A+B =$

(A) π (B) $\pi/2$

(C) $\pi/3$ **(D)** $\pi/4$

Solution:(Correct Answer:D)

(d) We know that $\sin(A+B) = \sin A\cos B + \cos A\sin B$

$= \frac{1}{\sqrt{10}}\sqrt{1-\frac{1}{5}} + \frac{1}{\sqrt{5}}\sqrt{1-\frac{1}{10}}$

$= \frac{1}{\sqrt{10}}\sqrt{\frac{4}{5}} + \frac{1}{\sqrt{5}}\sqrt{\frac{9}{10}}$

$= \frac{1}{\sqrt{50}}(2+3) = \frac{5}{\sqrt{50}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin(A+B) = \sin\frac{\pi}{4}$

Hence, $A+B = \frac{\pi}{4}$.

(4) If $\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+1012}\right)$
$-\left(\frac{1}{2\cdot1}+\frac{1}{4\cdot3}+\frac{1}{6\cdot5}+\ldots+\frac{1}{2024\cdot2023}\right)$
$= \frac{1}{2024}$, then α is equal to-

(A) 1367 (B) 1058

(C) 1056 **(D)** 1011

Solution:(Correct Answer:D)

$\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$ -
$\left\{\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\ldots+\left(\frac{1}{2023}-\frac{1}{2024}\right)\right\} = \frac{1}{2024}$

$\Rightarrow \left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$
$-\left\{\left(\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)+\ldots+\frac{1}{2023}\right.$
$\left.-\frac{1}{2024}-2\left(\frac{1}{2}+\frac{1}{4}+\ldots+\frac{1}{2022}\right)\right\} = \frac{1}{2024}$

$\Rightarrow \left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right) - \left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{2023}\right)$
$\frac{1}{2024}+\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{1011}\right) = \frac{1}{2024}$

$\Rightarrow \frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012} = \frac{1}{1012}+\frac{1}{1013}+\ldots+\frac{1}{2023}$

$\Rightarrow \alpha = 1011$

(5) Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}}+\beta x\right)^4, (1-3\beta x)^2$ and $\left(1-\frac{\beta}{2}x\right)^6, \beta > 0$, respectively form the first three terms of an $A.P.$ If d is the common difference of this $A.P.$, then $50-\frac{2d}{\beta^2}$ is equal to.

(A) 57 (B) 56

(C) 55 (D) 54

Solution:(Correct Answer:A)

${}^4C_2 \times \frac{\beta^2}{6}, -6\beta, -{}^6C_3 \times \frac{\beta^3}{8}$ are in A.P

$\beta^2 \frac{5}{2}\beta^3 = -12\beta$

$\beta = \frac{12}{5}$ or $\beta = -2 \therefore \beta = \frac{12}{5}$

$d = -\frac{72}{5} - \frac{144}{25} = -\frac{504}{25}$

$\therefore 50 - \frac{2d}{\beta^2} = 57$

(6) The equation $(a+b)^2 = 4ab\sin^2\theta$ is possible only when

(A) $2a = b$ **(B)** $a = b$

(C) $a = 2b$ (D) None of these

Solution:(Correct Answer:B)

(b) We have $(a+b)^2 = 4ab\sin^2\theta$

$\Rightarrow \sin^2\theta = \frac{(a+b)^2}{4ab} \leq 1$

$\Rightarrow (a+b)^2 - 4ab \leq 0$

$\Rightarrow (a-b)^2 \leq 0$

$\Rightarrow a = b.$

(7) The term independent of x in the expansion of $(1+x)^n\left(1+\frac{1}{x}\right)^n$ is

(A) $C_0^2 + 2C_1^2 + \ldots. + (n+1)C_n^2$

(B) $(C_0 + C_1 + \ldots. + C_n)^2$

(C) $C_0^2 + C_1^2 + \ldots.. + C_n^2$

(D) None of these

Solution:(Correct Answer:C)

(c) As in Previous question, obviously the term independent of x will be

${}^nC_0.{}^nC_0 + {}^nC_1.{}^nC_1 + \ldots..{}^nC_n.{}^nC_n = C_0^2 + C_1^2 + \ldots. + C_n^2.$

(8) If number of terms in the expansion of $(x-2y+3z)^n$ are 45, then $n =$

(A) 7 **(B)** 8

(C) 9 (D) None of these

Solution:(Correct Answer:B)

(b) $\frac{(n+1)(n+2)}{2} = 45$ or $n^2+3n-88 = 0 \Rightarrow n = 8$..

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(9) $x^5+10x^4a+40x^3a^2+80x^2a^3+80xa^4+32a^5=$

(A) $(x+a)^5$ (B) $(3x+a)^5$

(C) $(x+2a)^5$ (D) $(x+2a)^3$

Solution:(Correct Answer:C)

(c)
Conversely,$(x+a)^n = {}^nC_0+{}^nC_1x^{n-1}a+{}^nC_2x^{n-2}a^2+.....$
So,
$(x+2a)^5=x^5+10x^4a+40x^3a^2+80x^2a^3+80xa^4+32a^5.$

(10) Let the coefficient of x^r in the expansion of $(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+.....+(x+2)^{n-1}$ be α_r. If $\sum_{r=0}^{n}\alpha_r=\beta^n-\gamma^n, \beta,\gamma\in N$, then the value of $\beta^2+\gamma^2$ equals.....

(A) 23 (B) 24

(C) 20 **(D)** 25

Solution:(Correct Answer:D)

$(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+.....+(x+2)^{n-1}$
$\sum\alpha_r=4^{n-1}+4^{n-2}\times3+4^{n-3}\times3^2.....+3^{n-1}$
$=4^{n-1}\left[1+\frac{3}{4}+\left(\frac{3}{4}\right)^2.....+\left(\frac{3}{4}\right)^{n-1}\right]$
$=4^{n-1}\times\frac{1-\left(\frac{3}{4}\right)^n}{1-\frac{3}{4}}$
$=4^n-3^n=\beta^n-\gamma^n$
$\beta=4,\gamma=3$
$\beta^2+\gamma^2=16+9=25$

(11) If the arithmetic mean of two numbers be A and geometric mean be G, then the numbers will be

(A) $A\pm(A^2-G^2)$ (B) $\sqrt{A}\pm\sqrt{A^2-G^2}$

(C) $A\pm\sqrt{(A+G)(A-G)}$ (D) $\frac{A\pm\sqrt{(A+G)(A-G)}}{2}$

Solution:(Correct Answer:C)

(c) $A.M.=\frac{a+b}{2}=A$
and $G.M.=\sqrt{ab}=G$
On solving a and b are given by the values
$A\pm\sqrt{(A+G)(A-G)}$.
Trick : Let the numbers be $1,9$.
Then $A=5$ and $G=3$. Now put these values in options.
Here (c) $\Rightarrow 5\pm\sqrt{8\times2}$
$i.e.$ 9 and 1.

(12) $n^n\left(\frac{n+1}{2}\right)^{2n}$ is

(A) Less than $\left(\frac{n+1}{2}\right)^3$ (B) Greater than $\left(\frac{n+1}{2}\right)^3$

(C) Greater than $(n!)^3$ **(D)** (b) and (c) both

Solution:(Correct Answer:D)

(b) $y=n^n\left(\frac{n+1}{2}\right)^{2n}$
Put $n=2$, $y=2^2\left(\frac{3}{2}\right)^4=4.\frac{81}{8\times2}=\frac{81}{4}\tilde{-}20$
Option $(a)=\left(\frac{n+1}{2}\right)^3=\frac{27}{8}<y$
Option $(b)=\left(\frac{n+1}{2}\right)^3=\frac{27}{8}<y$
Option $(c)=(2!)^3=8<y$

(13) Let $a,b,c>1, a^3,b^3$ and c^3 be in $A.P.$, and $\log_a b$, $\log_c a$ and $\log_b c$ be in G.P. If the sum of first 20 terms of an $A.P.$, whose first term is $\frac{a+4b+c}{3}$ and the common difference is $\frac{a-8b+c}{10}$ is -444, then abc is equal to

(A) 343 **(B)** 216

(C) $\frac{343}{8}$ (D) $\frac{125}{8}$

Solution:(Correct Answer:B)

As a^3,b^3,c^3 be in $A.P.\rightarrow a^3+c^3=2b^3$ $\log_a^b,\log_c^a,\log_b^c$ are in $G.P.$
$\therefore\frac{\log b}{\log a}\cdot\frac{\log c}{\log b}=\left(\frac{\log a}{\log c}\right)^2$
$\therefore(\log a)^3=(\log c)^3\Rightarrow a=c$
From (1) and (2)
$a=b=c$
$T_1=\frac{a+4b+c}{3}=2a; d=\frac{a-8b+c}{10}=\frac{-6a}{10}=\frac{-3}{5}a$
$\therefore S_{20}=\frac{20}{2}\left[4a+19\left(-\frac{3}{5}a\right)\right]$
$=10\left[\frac{20a-57a}{5}\right]$
$=-74a$
$\therefore -74a=-444\Rightarrow a=6$
$\therefore abc=6^3=216$

(14) Fifth term of a $G.P.$ is 2, then the product of its 9 terms is

(A) 256 **(B)** 512

(C) 1024 (D) None of these

Solution:(Correct Answer:B)

(b) Let the 9 terms of a $G.P.$ is
$\frac{a}{r^4},\frac{a}{r^3},\frac{a}{r^2},\frac{a}{r},a,ar,ar^2,ar^3,ar^4.$
Given, fifth term $a=2$
Hence product of 9 terms is $a^9=(2)^9=512.$

(15) If $p=\frac{2\sin\theta}{1+\cos\theta+\sin\theta}$, and $q=\frac{\cos\theta}{1+\sin\theta}$, then

(A) $pq=1$ (B) $\frac{q}{p}=1$

(C) $q-p=1$ **(D)** $q+p=1$

Solution:(Correct Answer:D)

(d) $p=\frac{2\sin\theta}{1+\cos\theta+\sin\theta},\ q=\frac{\cos\theta}{1+\sin\theta}$
$\Rightarrow p+q=\frac{\cos\theta}{1+\sin\theta}+\frac{2\sin\theta}{1+\sin\theta+\cos\theta}$
$\Rightarrow p+q=1.$

(16) The value of $36(4\cos^2 9^\circ-1)(4\cos^2 27^\circ-1)(4\cos^2 81^\circ-1)(4\cos^2 243^\circ-1)$ is

(A) 54 (B) 18

(C) 27 **(D)** 36

Solution:(Correct Answer:D)

$4\cos^2\theta-1=4\left(1-\sin^2\theta\right)-1=3-4\sin^2\theta=\frac{\sin3\theta}{\sin\theta}$
so given expression can be written as
$36\times\frac{\sin27^\circ}{\sin9^\circ}\times\frac{\sin81^\circ}{\sin27^\circ}\times\frac{\sin243^\circ}{\sin81^\circ}\times\frac{\sin729^\circ}{\sin243^\circ}$
$36\times\frac{\sin729^\circ}{\sin9^\circ}=36$

(17) The sum of all two digit numbers which, when divided by 4, yield unity as a remainder is

(A) 1190 (B) 1197

(C) 1210 (D) None of these

Solution:(Correct Answer:C)

(c) The given numbers are $13,17,.....97.$
This is an AP with first term 13 and common difference 4.
Let the number of terms be n.
Then $97=13+(n-1)4$
$\Rightarrow 4n=88$
$\Rightarrow n=22$
Therefore the sum of the numbers
$=\frac{22}{2}[13+97]=11(110)=1210.$

(18) If $T_0,T_1,T_2,....T_n$ represent the terms in the expansion of $(x+a)^n$, then $(T_0-T_2+T_4-....)^2+(T_1-T_3+T_5-....)^2=$

(A) (x^2+a^2) **(B)** $(x^2+a^2)^n$

(C) $(x^2+a^2)^{1/n}$ (D) $(x^2+a^2)^{-1/n}$

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Solution:(Correct Answer:B)

(b) From the given condition, replacing a by ai and -ai respectively,
we
get$(x+ai)^n = (T_0 - T_2 + T_4 - ....) + i(T_1 - T_3 + T_5 - ....)$ ....(i)
and $(x-ai)^n = (T_0 - T_2 + T_4 - ....) - i(T_1 - T_3 + T_5 - ....)$ .....(ii)
Multiplying (ii) and (i) we get required result i.e.
$(x^2+a^2)^n = (T_0 - T_2 + T_4 - ....)^2 + (T_1 - T_3 + T_5 - ....)^2$

(19) $2\cos^2\theta - 2\sin^2\theta = 1$, then $\theta = ...^o$

(A) 15 **(B)** 30

(C) 45 (D) 60

Solution:(Correct Answer:B)

(b) $2\cos^2\theta - 2\sin^2\theta = 1$
== > $2\cos 2\theta = 1$
== > $\cos 2\theta = \frac{1}{2} = \cos 60^o$
== > $2\theta = 60^o$
$\Rightarrow \theta = 30^o$.

(20) $2+4+7+11+16+$ .....to n terms =

(A) $\frac{1}{6}(n^2+3n+8)$ **(B)** $\frac{n}{6}(n^2+3n+8)$

(C) $\frac{1}{6}(n^2-3n+8)$ (D) $\frac{n}{6}(n^2-3n+8)$

Solution:(Correct Answer:B)

(b) We have $S = 2+4+7+11+16+ ..... + T_n$
Again $S = 2+4+7+11+ ..... + T_{n-1} + T_n$
Subtracting, we get
$0 = 2 + \{2+3+4+5+ .....(T_n - T_{n-1})\} - T_n$
$T_n = 2 + \frac{1}{2}(n-1)(4+\{n-2)1\} = \frac{1}{2}(n^2+n+2)$
Now $S = \Sigma T_n = \frac{1}{2}\Sigma(n^2+n+2)$
$= \frac{1}{2}(\Sigma n^2 + \Sigma n + 2\Sigma 1)$
$= \frac{1}{2}\left\{\frac{1}{6}n(n+1)(2n+1) + \frac{1}{2}n(n+1) + 2n\right\}$
$= \frac{n}{12}\{(n+1)(2n+1+3)+12\}$
= $\frac{n}{6}\{(n+1)(n+2)+6\}$
$= \frac{n}{6}(n^2+3n+8)$.

Mathematics - Section B (NUMERIC)

(21) The coefficient of x in the expansion of $[\sqrt{1+x^2} - x]^{-1}$ in ascending powers of x, when $|x| < 1$, is

Solution:

(d) $[\sqrt{1+x^2} - x]^{-1} = \frac{1}{\sqrt{1+x^2}-x} \times \frac{(\sqrt{1+x^2}+x)}{(\sqrt{1+x^2}+x)}$
$= \frac{\sqrt{1+x^2}+x}{1+x^2-x^2} = x + \sqrt{1+x^2} = x + (1+x^2)^{1/2}$
$= x + 1 + \frac{1}{2}x^2 + \frac{1}{2}\left(-\frac{1}{2}\right)\frac{x^4}{2!} + ...$
Obviously, coefficient of x is 1.

(22) Let $75...57$ denote the $(r+2)$ digit number where the first and the last digits are 7 and the remaining r digits are 5. Consider the sum $S = 77 + 757 + 7557 + ... + 75....57$. If $S = \frac{75^{98}...57+m}{n}$, where m and n are natural numbers less than 3000, then the value of $m+n$ is

Solution:

$S = 77 + 757 + 7557 + ... + 75.....57$
98
$10S = \quad 770 + 7570 + ... + 75...570 + 755....570$
$9\,S = -77 + \underbrace{13 + 13 + ... + 13}_{98 \text{ tims}} + 75.....80$
$= -77 + 13 \times 98 + 75.....7 + 13$
S $= \frac{75....57+1210}{9}$
m $= 1210$
n $= 9$
m + n $= 1219$

(23) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ =$

Solution:

(c) $\tan\ 9^o - \tan\ 27^o - \tan\ 63^o + \tan\ 81^o$
$= \tan\ 9^o - \tan\ 27^o - \cot\ 27^o + \cot\ 9^o$
$= (\tan\ 9^o + \cot\ 9^o) - (\tan\ 27^o + \cot\ 27^o)$
$= \frac{\cos(9^o-9^o)}{\sin 9^o \cos 9^o} - \frac{\cos(27^o-27^o)}{\sin 27^o.\cos 27^o} = \frac{2}{\sin 18^o} - \frac{2}{\sin 54^o}$
$= 2\left\{\frac{\sin\ 54^o - \sin\ 18^o}{\sin\ 18^o\ \sin\ 54^o}\right\}$
$= 2.\ \frac{2.\ \cos\ 36^o.\ \sin\ 18^o}{\sin\ 18^o.\ \sin\ 54^o} = 4$

(24) Consider a sequence whose sum of first n -terms is given by $S_n = 4n^2 + 6n, n \in N$, then T_{15} of this sequence is -

Solution:

$T_n = S_n - S_{n-1}$
$\Rightarrow T_n = (4n^2+6n) - (4(n-1)^2 + 6(n-1))$
$T_n = 8n + 2 \Rightarrow T_{15} = 122$

(25) If $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + ...$, then 160 S is equal to..... .

Solution:

S $= \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + ...$
$\frac{1}{5}$ S $= \frac{7}{5^2} + \frac{9}{5^3} + \frac{13}{5^4} + ...$
On subtracting
$\frac{4}{5}S = \frac{7}{5} + \frac{2}{5^2} + \frac{4}{5^3} + \frac{6}{5^4} + ...$
$S = \frac{7}{4} + \frac{1}{10}\left(1 + \frac{2}{5} + \frac{3}{5^2} + ...\right)$
$S = \frac{7}{4} + \frac{1}{10}\left(1 - \frac{1}{5}\right)^{-2}$
$= \frac{7}{4} + \frac{1}{10} \times \frac{25}{16} = \frac{61}{32}$
$\Rightarrow 160$ S $= 5 \times 61 = 305$

Physics - Section A (MCQ)

(26) A pendulum clock loses 12 sec a day if the temperature is $40°C$ and gains 4 sec a day if the temperature is $20°C$. The temperature at ...$°C$ clock will show correct time.

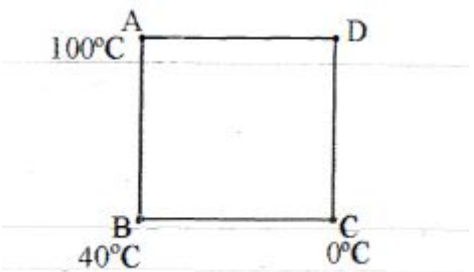
(A) 25 (B) 60

(C) 30 (D) 55

Solution:(Correct Answer:A)

$\Delta T = \frac{1}{2} \propto (\Delta\theta)T;$
$12 = \frac{1}{2} \propto (40-\theta) \times T ...$ (i)
$4 = \frac{1}{2}\alpha(\theta - 20)T ...$ (ii)
$(i)/(ii)$
$\frac{12}{4} = \frac{40-\theta}{\theta-20}$
$\Rightarrow 3\theta - 60 = 40 - \theta$
$\Rightarrow 4\theta = 100$
$\Rightarrow \theta = 25°C$

(27) Four conducting rods are joined to make a square. All rods are identical and ends A, B and C are maintained at given temperatures. choose $INCORRECT$ statement for given arrangement in steady state. (value of $\frac{KA}{L}$ is $1\frac{J}{S°C}$, symbols , have their usual meaning)



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(A) Heat current in AB is equal to 1.5 times of heat current in BC

(B) Temperature of end D is $50\,^oC$

(C) Heat current in AB is equal to heat current in BC

(D) Heat current withdrawn at end B is $20\ J/S$

Solution:(Correct Answer:C)

$i_{AB} = \frac{KA}{L}(100-40), i_{BC} = \frac{KA}{L}(40-0)$
$i_{AD} - i_{DC}$ so $T_D = \frac{T_A+T_C}{2} = \frac{100+0}{2} = 50°C$

(28) Two mercury drops (each of radius r) merge to form a bigger drop. The surface energy of the bigger drop, if T is the surface tension is

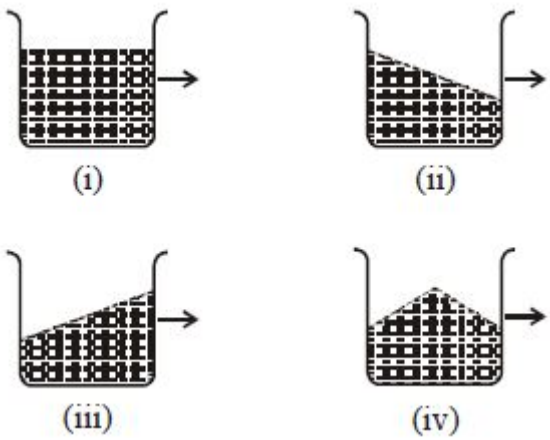
(A) $2^{5/3}\pi r^2 T$ (B) $4\pi r^2 T$

(C) $2\pi r^2 T$ **(D)** $2^{8/3}\pi r^2 T$

Solution:(Correct Answer:D)

$E = 4\pi R^2 T\ [R = n\frac{1}{3}r]$

(29) A vessel containing water is moving with a constant speed towards right along a straight horizontal path. Which of the following diagrams represents the surface of liquid?



(A) (i) (B) (ii)

(C) (iii) (D) (iv)

Solution:(Correct Answer:A)

As the vessel containing water is moving with a constant speed, thus no pseudo force acts on the water in the vessel frame. Also the vessel frame is an inertial frame for water. Hence the moving liquid (water) forms the surface as same as that when the liquid was at rest.
Hence option A is correct.

(30) A soap film of surface tension $3 \times 10^{-2}\ Nm^{-1}$ formed in rectangular frame, can support a straw. The length of the film is $10\,cm$. Mass of the straw the film can support is ..... gm

(A) 0.06 **(B)** 0.6

(C) 6 (D) 60

Solution:(Correct Answer:B)

(b) The weight of straw will be balanced by the force of surface tension
$mg = 2Tl$== > $m = \frac{2Tl}{g}$
$= \frac{2\times3\times10^{-2}\times10\times10^{-2}}{9.8}kg = 0.6gm$

(31) In a test experiment on a model aeroplane in a wind tunnel, the flow speeds on the upper and lower surfaces of the wing are $70\ ms^{-1}$ and $63\ ms^{-1}$ respectively. What is the lift on the wing if its area is $2.5\ m^2$? Take the density of atr to be $1.3\ kgm^{-3}$.

(A) $1.51 \times 10^3\ N$ (B) $3.64 \times 10^4\ N$

(C) $2.67 \times 10^5\ N$ (D) $8.60 \times 10^2\ N$

Solution:(Correct Answer:A)

Speed of wind on the upper surface of the wing, $V_1 = 70m/s$ Speed of wind on the lower surface of the wing, $V_2 = 63m/s$ Area of the wing, $A = 2.5m^2$ Density of air, $\rho = 1.3kgm^{-3}$
According to Bernoulli's theorem, we have the relation:
$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$
$P_2 - P_1 = \frac{1}{2}\rho\left(V_1^2 - V_2^2\right)$
Where,
$P_1 =$ Pressure on the upper surface of the wing
$P_2 =$ Pressure on the lower surface of the wing
The pressure difference between the upper and lower surfaces of the wing provides lift to the aeroplane.
Lift on the wing $= (P_2 - P_1)\,A$
$= \frac{1}{2}\rho\left(V_1^2 - V_2^2\right)A$
$= \frac{1}{2}1.3\left((70)^2 - (63)^2\right) \times 2.5$
$= 1512.87$
$= 1.51 \times 10^3 N$
Therefore, the lift on the wing of the aeroplane is $1.51 \times 10^3\ N$.

(32) A metallic ball and highly stretched spring are made of the same material and have the same mass. They are heated so that they melt, the latent heat required

(A) Are the same for both

(B) Is greater for the ball

(C) Is greater for the spring

(D) For the two may or may not be the same depending upon the metal

Solution:(Correct Answer:A)

(a)Latent heat is independent of configuration. Ordered energy spent in stretching the spring will not contribute to heat which is disordered kinetic energy of molecules of substance.

(33) When heat energy of $1500\ Joules$, is supplied to a gas at constant pressure $2.1 \times 10^5\ N/m^2$, there was an increase in its volume equal to $2.5 \times 10^{-3}\ m^3$. The increase in internal energy of the gas in Joules is ..... J

(A) 450 (B) 525

(C) 975 (D) 2025

Solution:(Correct Answer:C)

(c) According to FLOT
$\Delta Q = \Delta U + P(\Delta V)$== > $\Delta U = \Delta Q - P(\Delta V)$
$= 1500 - (2.1 \times 10^5)(2.5 \times 10^{-3}) = 975\ Joule$

(34) What will be the work done in increasing the radius of soap bubble from $\frac{r}{2}$ to $2r$, if surface tension of soap solution is T ?

(A) $3\pi r^2 T$ **(B)** $15\pi r^2 T$

(C) $9\pi r^2 T$ (D) $12\pi r^2 T$

Solution:(Correct Answer:B)

(b)
Surface energy of soap bubble :
$U = T \times 4\pi r^2$
$U_1 = T \times 4\pi(\frac{r}{2})^2 = T \times 4\pi(\frac{r^2}{4}) = \pi T r^2$
$U_2 = T \times 4\pi(2r)^2 = 16\pi T r^2$
The work done in increasing the radius of soap bubble is equal to the increase in surface energy.

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$\therefore W = (U_2 - U_1)$
$= 16\pi Tr^2 - \pi Tr^2$
$= 15\pi Tr^2$
Thus, the work done in increasing the radius of soap bubble will be $= 15\pi Tr^2$

(35) The ratio of radiant energies radiated per unit surface area by two bodies is $16:1$, the temperature of hotter body is $1000K$, then the temperature of colder body will be ..... K

(A) 250 **(B)** 500

(C) 1000 (D) 62.5

Solution:(Correct Answer:B)

(b) $Q \propto T^4$
== > $\frac{Q_1}{Q_2} = \frac{T_1^4}{T_2^4}$
== > $T_2^4 = \left(\frac{E_2}{E_1}\right) T_1^4$
== > $T_2^4 = \frac{1}{16} \times (1000)^4 = \left(\frac{1000}{2}\right)^4$
== > $T_2 = 500K$

(36) A cube of wood supporting $200\,gm$ mass just floats in water. When the mass is removed, the cube rises by $2\,cm$. ..... cm is the side of cube .

(A) 5 **(B)** 10

(C) 15 (D) 20

Solution:(Correct Answer:B)

weight of mass $=$ weight of water displaced
$=$ V $\times \rho_{\text{w}} \Rightarrow$ V $= \frac{200}{1} = 200$cc
$\therefore \quad 2 \times a \times a = 200 \Rightarrow a = 10$cm

(37) An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.
A. The work done by gas during the process is zero.
B. The heat added to gas is different from change in its internal energy.
C. The volume of the gas is increased.
D. The internal energy of the gas is increased.
E. The process is isochoric (constant volume process)
Choose the correct answer from the options given below : $-$

(A) A, B, C, D Only **(B)** A, D, E Only

(C) E Only (D) A, C Only

Solution:(Correct Answer:B)

Given that
P $=$ kT
$\frac{\text{P}}{\text{T}} =$ constant
$\therefore$ Volume is constant or isochoric process.
$\therefore$ $\text{W}_{\text{D}} = 0$
$\therefore$ Q $= \Delta$U
Also temperature increases hence internal energy increases.

(38) A black body at $200K$ is found to exit maximum energy at a wavelength of $14\mu m$. When its temperature is raised to $1000K$, the wavelength at which maximum energy is emitted is

(A) $14\mu m$ (B) $70\mu F$

(C) $2.8\mu m$ (D) $2.8mm$

Solution:(Correct Answer:C)

(c) $\lambda_{m_1} T_1 = \lambda_{m_2} T_2 \Rightarrow \lambda_{m_2} = \frac{\lambda_{m_1} T_1}{T_2} = \frac{14 \times 200}{1000} = 2.8\ \mu m$

(39) In rainy season, on a clear night the black seat of a bicycle becomes wet because

(A) It absorbs water vapour

(B) Black seat is good absorber of heat

(C) Black seat is good radiator of heat energy

(D) None of the above

Solution:(Correct Answer:C)

In the rainy season, on a clear night, the black seat of a bicycle becomes wet because black seat is a good radiator of heat energy.

(40) A liquid drop of diameter D breaks into 27 tiny drops. The change in energy is

(A) $2\pi TD^2$ (B) $4\pi TD^2$

(C) πTD^2 (D) None of these

Solution:(Correct Answer:A)

$\Delta SE = 4\pi R^2 T\ (n^{1/3} - 1)$
$= 4\pi \frac{D^2}{4} T[(27)^{1/3} - 1] = 2\pi D^2 T$

(41) For an isothermal expansion of a perfect gas, the value of $\frac{\Delta P}{P}$ is equal

(A) $-\gamma^{1/2} \frac{\Delta V}{V}$ **(B)** $-\frac{\Delta V}{V}$

(C) $-\gamma \frac{\Delta V}{V}$ (D) $-\gamma^2 \frac{\Delta V}{V}$

Solution:(Correct Answer:B)

(b)Differentiate $PV =$constant w.r.t V
== > $P\Delta V + V\Delta P = 0$== > $\frac{\Delta P}{P} = -\frac{\Delta V}{V}$

(42) In supplying 400 calories of heat to a system, the work done will be

(A) 400 joules **(B)** 1672 joules

(C) 1672 watts (D) 1672 ergs

Solution:(Correct Answer:B)

(b) $W = JQ = 4.18 \times 400 = 1672\,joule$

(43) A container of height $10\,cm$ is filled with water. There is a hole at bottom. Find the pressure difference between points at top and bottom.

(A) $1000\,Pa$ (B) zero

(C) $1\,Pa$ (D) $100\,Pa$

Solution:(Correct Answer:A)

The difference in the pressure at the two points is calculated as,
$P_A - P_B = \rho gh$
$= 10^3 \times 10 \times 10 \times 10^{-2} Pa$
$= 1000Pa$

(44) Assertion :- A brass disc is fast fitted in a hole in a steel plate. The system must be cooled to loosen the disc from the hole.
Reason :- The coefficient of linear expansion of brass, is greater than the coefficient of linear expansion of steel.

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Solution:(Correct Answer:A)

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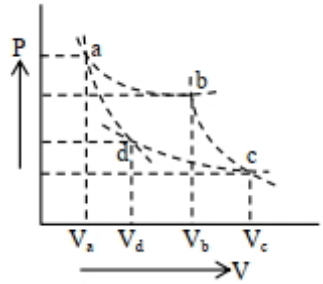
(45) Two different adiabatic paths for the same gas intersect two isothermal curves as shown in $P-V$ diagram. The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:



(A) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^{-1}$ (B) $\frac{V_a}{V_d}\neq\frac{V_b}{V_c}$

(C) $\frac{V_a}{V_d}=\frac{V_b}{V_c}$ (D) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^{2}$

Solution:(Correct Answer:C)

For adiabatic process

$TV^{\gamma-1}=$ constant

$T_a\cdot V_a^{\gamma-1}=T_d\cdot V_d^{\gamma-1}$

$\left(\frac{V_a}{V_d}\right)^{\gamma-1}=\frac{T_d}{T_a}$

$T_b\cdot V_b^{\gamma-1}=T_c\cdot V_c^{\gamma-1}$

$\left(\frac{V_b}{V_c}\right)^{\gamma-1}=\frac{T_c}{T_b}$

$T_b\cdot V_b^{\gamma-1}=T_c\cdot V_c^{\gamma-1}$

$\left(\frac{V_b}{V_c}\right)^{\gamma-1}=\frac{T_c}{T_b}$

$\frac{V_a}{V_d}=\frac{V_b}{V_c}\quad\left(\begin{array}{c}\because T_d=T_c\\ T_a=T_b\end{array}\right)$

Physics - Section B (NUMERIC)

(46) A ball whose density is $0.4\times10^3\,kg/m^3$ falls into water from a height of $9\,cm$. To what depth does the ball sink ? ..... cm

Solution:

energy of the ball at the surface of the water

$=mgh=V\rho_b gh$

upthrust on ball $=V\rho_w g$

net force on the ball in water $=(\rho_w-\rho_b)Vg$

it is constant and upwards

let it sink upto depth d then

$(\rho_m-\rho_b)V_\rho d=V\rho_b gh$

$\Rightarrow d=\frac{\rho_b}{\rho_w-\rho_b}h$

$\Rightarrow d=\frac{0.4\times10^3}{1\times10^3-0.4\times10^3}\times9\text{cm}=\frac{4}{6}\times9=6\text{cm}$

(47) A big drop is formed by coalescing 1000 small identical drops of water. If E_1 be the total surface energy of 1000 small drops of water and E_2 be the surface energy of single big drop of water, the E_1 : E_2 is x : 1 where x =_____.

Solution:

$\rho\left(\frac{4}{3}\pi r^3\right)1000=\frac{4}{3}\pi R^3\rho$

$R=10r$

$E_1=1000\times4\pi r^2\times S$

$E_2=4\pi(10r)^2S$

$\frac{E_1}{E_2}=\frac{10}{1},x=10$

(48) If heat is added at constant volume, $6300\ J$ of heat are required to raise the temperature of an ideal gas by $150\ K$. If instead, heat is added at constant pressure, 8800 joules are required for the same temperature change. When the temperature of the gas changes by $300\ K$, the internal energy of the gas changes by ..... J

Solution:

Heat added at constant volume is equal to the change in internal energy of the system. Change in internal energy

$\Delta U=nC_v\Delta T$

For $\Delta T=150K,\quad \Delta U=Q_v=6300\text{J}$

$\therefore\quad 6300=nC_v(150)\ldots(1)$

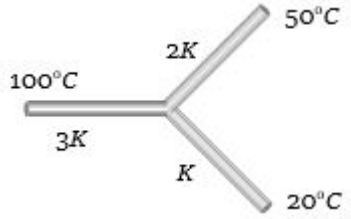
Let the change in internal energy be ΔU for $\Delta T=300K$

$\therefore\Delta U=nC_v(300)\ldots(2)$

Dividing (2) and (1) we get $\frac{\Delta U}{6300}=\frac{300}{150}$

$\Longrightarrow\Delta U=12600\text{J}$

(49) Three rods of the same dimension have thermal conductivities $3K$, $2K$ and K . They are arranged as shown in fig. Given below, with their ends at $100^oC, 50^oC$and 20^oC. The temperature of their junction is ..... ^oC



Solution:

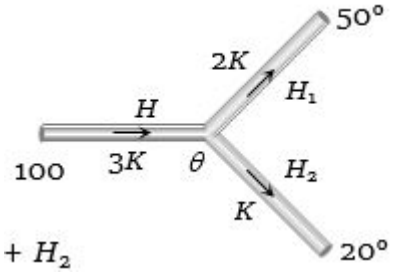
(b) Let the temperature of junction be θ then according to following figure.

$H=H_1+H_2$

== > $\frac{3K\times A\times(100-\theta)}{l}=\frac{2KA(\theta-50)}{l}+\frac{KA(\theta-20)}{l}$

== > $300-3\theta=3\theta-120$

== > $\theta=70°C$



(50) A faulty thermometer has its fixed points marked 5° and 95°. This thermometer reads the temperature of a body as 59°. Then the correct temperature of the body on celsius scale is ..... ^{o}

Solution:

If T_C is the correct temperature on celsisud scale, then

$\frac{T_C-0}{100}=\frac{x\text{ -lower fixed point}}{\text{upper fixed point - lower fixxed point}}$

$\frac{T_C}{100}=\frac{59-5}{95-5}=\frac{540}{90}=\frac{3}{5}$

$T_C=\frac{3000}{5}=60°$

Chemistry - Section A (MCQ)

(51) Incorrect order is

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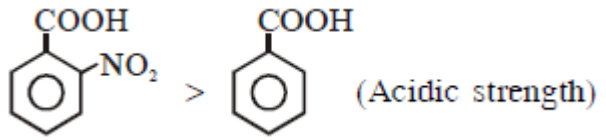
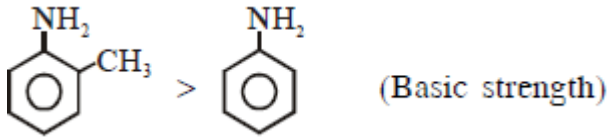
(A)

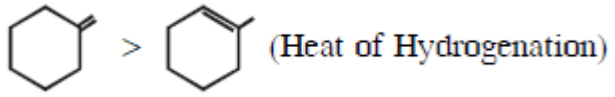


(B)

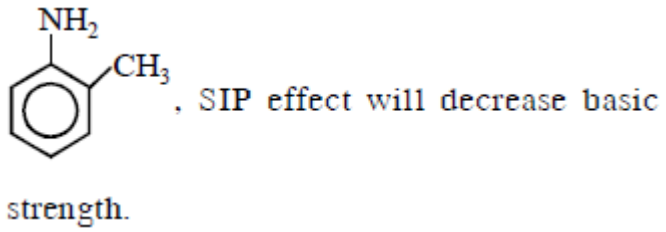


(C) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-O^{\ominus} > CH_3-\overset{\ominus}{C}H_2$ (Stability)

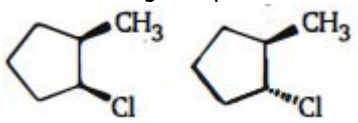
(D)



Solution:(Correct Answer:B)



(52) The following compounds are identical with respect to



(A) molecular composition (B) boiling point

(C) melting point (D) $IUPAC$ name

Solution:(Correct Answer:A)

(a) Both molecule has same molecular formula $C_6H_{11}Cl$ so have same molecular wt. and composition because both have substituents on same atoms.

(53) In the given molecule $(NC)_3CCCCHCHBr$ number of σ and $\pi-$ bonds respectively are

(A) 13 and 9 (B) 12 and 8

(C) 7 and 3 (D) 10 and 5

Solution:(Correct Answer:A)

(54) In the sulphonation, acetylation and formylation of benzene the group of effective electrophiles would be :

(A) $SO_3^+, CH_3 \equiv \overset{+}{O}, H\overset{+}{C}O$

(B) $SO_3, CH_3-C \equiv \overset{+}{O}, H\overset{+}{C}O$

(C) $SO_3, CH_3CHO, CO+HCl$

(D) HSO_3, CH_3CO, HCO

Solution:(Correct Answer:B)

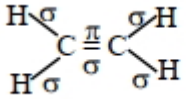
(55) Number of σ and π bonds present in ethylene molecule is respectively :

(A) 3 and 1 (B) 5 and 2

(C) 4 and 1 **(D)** 5 and 1

Solution:(Correct Answer:D)

1π bond.

ethylene is 

, it has 5σ bonds and

(56) Which of the following compounds will show geometrical isomerism

(A) $2-$ Butene (B) Propene

(C) Ethylene (D) $2-$ Methyl $-2-$ butene

Solution:(Correct Answer:A)

(57) Geometrical isomerism is shown by

(A) $2-$butene (B) $2-$butyne

(C) $2-$butanol (D) Butanal

Solution:(Correct Answer:A)

(a) $\underset{\underset{CH_3}{|}}{HC}=\underset{\underset{CH_3}{|}}{CH}$ and $\underset{\underset{CH_3}{|}}{HC}=\overset{\overset{CH_3}{|}}{CH}$

(Cis) 2−butene (Trans) 2−butene

(58) The halogen which is most reactive in the halogenation of alkenes under sunlight is

(A) Chlorine (B) Bromine

(C) Iodine (D) All equal

Solution:(Correct Answer:A)

(a)$C_2H_4+Cl_2 \xrightarrow{CCl_4} \underset{\text{EtheleneChloride}}{Cl-CH_2-CH_2-Cl}$

(59) Correctly matched products are

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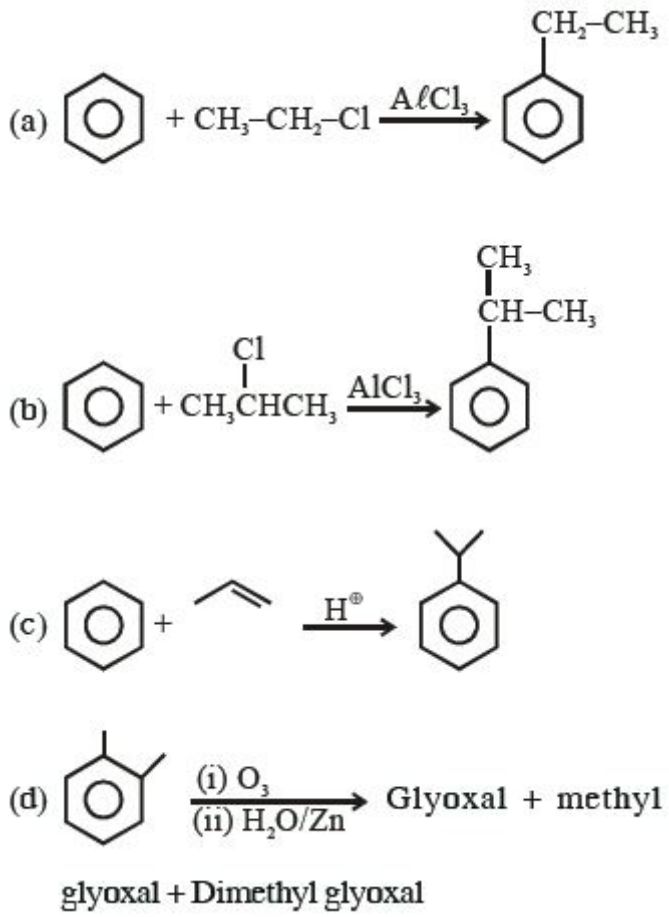
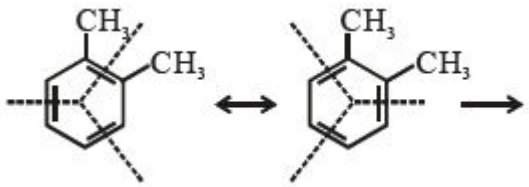


(A) b, c (B) b, c, d

(C) a, b, c, d (D) a, b, d

Solution:(Correct Answer:C)



(60) The $IUPAC$ name of $\underset{COOH}{CH_2} - CH_2 - \underset{COOH}{CH_2}$ is

(A) Pentane $-1,5-$ dioic acid

(B) Pentane $-1,5-$ dicarboxylic acid

(C) Propane $-1,3-$ dioic acid

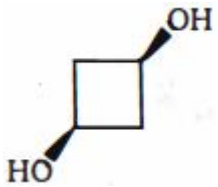
(D) none of these

Solution:(Correct Answer:A)

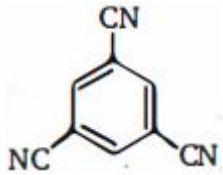
The IUPAC name of the following compound is pentane-$1,5$-dioic acid.

(61) Which one of the following compounds has non zero dipole moment?

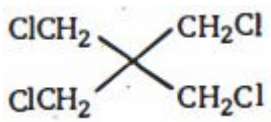
(A)



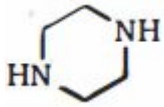
(B)



(C)



(D)



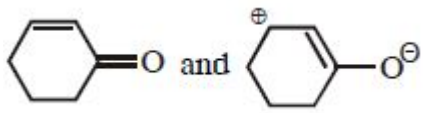
Solution:(Correct Answer:A)

(a) Cis molecule has non-zero dipole moment

(62) Which of the following pairs does not represent resonating structures ?

(A) $CH_3 - \overset{O}{\overset{||}{C}} - CH_3$ and $CH_3 - \overset{OH}{\overset{|}{C}} = CH_2$

(B)



(C) $CH_3 - \overset{\oplus}{\overset{O}{\overset{||}{C}}} - H$ and $CH_3 - \underset{\oplus}{\overset{OH}{\overset{|}{C}}} - CH_3$

(D) $CH_2 = C = O$ and $\overset{\ominus}{C} H_2 - C \equiv \overset{\oplus}{O}$

Solution:(Correct Answer:A)

$CH_3 - \overset{O}{\overset{||}{C}} - CH_3$ and $CH_3 - \overset{OH}{\overset{|}{C}} = CH_2$ are tautomers

(63) The structure of $4-$ methyl pentene -2 is

(A) $(CH_3)_2CH - CH_2CH = CH_2$

(B) $(CH_3)_2CH - CH = CH - CH_3$

(C) $(CH_3)_2CH - CH_2CH = CH - CH_3$

(D) $(CH_3)_2C = CHCH_2CH_3$

Solution:(Correct Answer:B)

The structure of 4 -methyl pent-2-ene is $(CH_3)_2 CH - CH = CH - CH_3$.
$(CH_3)_2 CH - CH_2CH = CH_2$ is named as 4-methyl pent-1-ene.
$(CH_3)_2 CH - CH_2 - CH = CH - CH_3$ is named as 5-methyl hex-2-ene.
$(CH_3)_2 C = CH - CH_2 - CH_3$ is named as 4-methyl pent-3-ene.

(64) Which is an electrophile

(A) $AlCl_3$ (B) CN^-

(C) NH_3 (D) CH_3OH

Solution:(Correct Answer:A)

(a) $AlCl_3$ is lewis acid i.e., electron deficient compound. So it is electrophile.

(65) Which is the most stable conformer along the $2,3\,C - C$ bond axis of the compound ?

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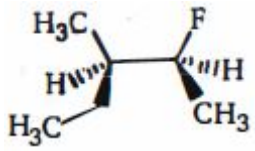
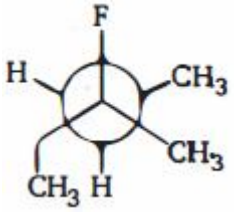
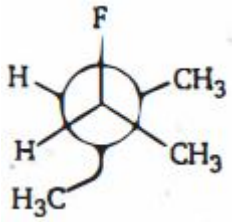


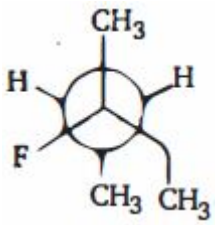
(A)



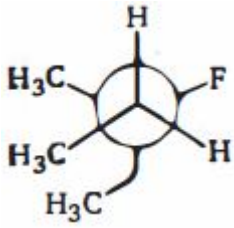
(B)



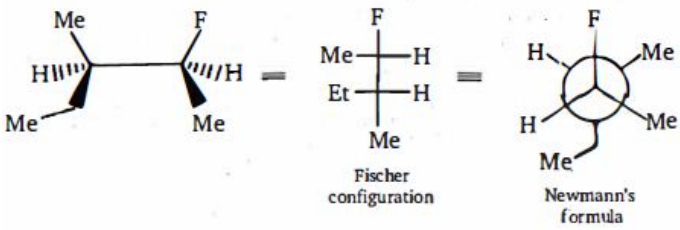
(C)



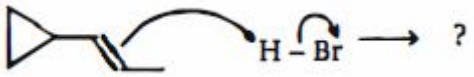
(D)



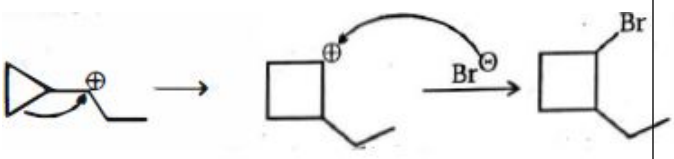
Solution:(Correct Answer:B)



(66) Taking into account the stability of various carbocations and, as well as the rules governing mechanisms of carbocation rearrangements, which reaction is most likely to occur during the given reaction ?



(A)



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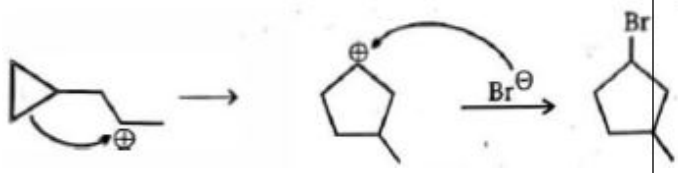
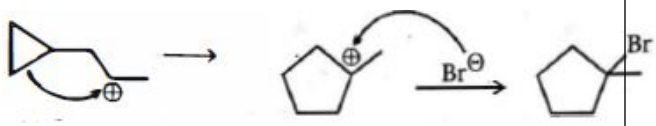


(C)



(D) None

Solution:(Correct Answer:D)

This carbocatioon rearrangement is not favourable.

None

(σ-resonance stabilized)

(67) The $IUPAC$ name of $\underset{CH_3}{CH_3-CH}-\underset{O}{\underset{\|}{C}}-\underset{O}{\underset{\|}{C}}-\underset{Br}{NH}$ is

(A) $(N-$ Bromo $)-2-$ keto $-3-$ methylbutanamide

(B) $(N-$ Bromo $)-2-$ keto $-4-$ methylbutanamide

(C) $(N-$ Bromo $)-1,2-$ diketo $-3-$ methylbutanamine carboxamide

(D) $(N-$ Bromo$)-1-$ keto $-2-$ methylpropane

Solution:(Correct Answer:A)

This is a N - bromo acidamide compound having a keto group. IUPAC name:- $(N-$ Bromo $)$-2- keto -3 - methylbutanamide.

(68) Minimum number of carbon atoms required for compound to show positional isomerism ?

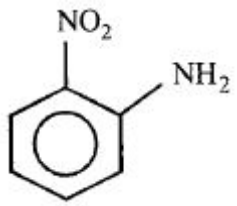
(A) 4 (B) 3

(C) 2 (D) 1

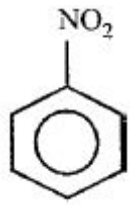
Solution:(Correct Answer:C)

(69) Which compound exhibits maximum dipole moment among the following ?

(A)

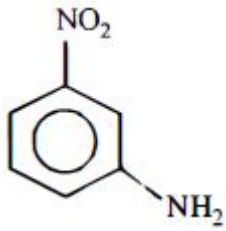


(B)



(C)

(D)



Solution:(Correct Answer:C)

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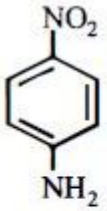
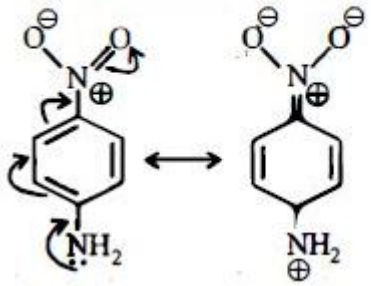


Dipole moment = (Distance between opposite charges) × (charge, q)

$$\mu = q \times d$$

So, greater the distance between the opposite charges higher the dipole. Due to

the resonance the greater charge separation occurs between charges due to linearity.



(70) Amongst the following, the compound that can be most readily sulphonated is

(A) Benzene (B) Nitrobenzene

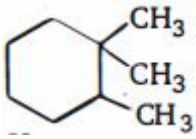
(C) Toluene (D) Chlorobenzene

Solution:(Correct Answer:C)

Toulene can be most readily sulfonated because of the electron releasing group attached to it. Nucleophilicity increases resulting in a fast attack to $E^{\oplus}$ (during sulphonation).

Chemistry - Section B (NUMERIC)

(71) How many geometrical isomers are possible for the above compound ?



(72) How many secondary carbon atoms does methyl cyclopropane have ?

Solution:

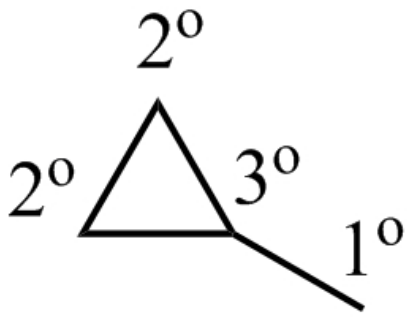
Methyl cyclopropane has two secondary carbon atoms, one primary carbon atom and one tertiary carbon atom.

A primary carbon atom is attached to only one other carbon atom.

A secondary carbon atom is attached to two other carbon atoms.

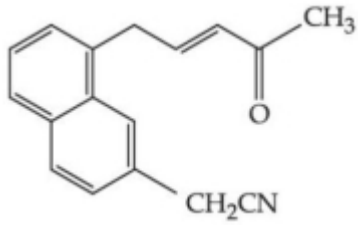
A tertiary carbon atom is attached to three other carbon atoms

A quaternary carbon atom is attached to four other carbon atoms.



methyl cyclopropane

(73) Number of electrophilic centre in the given compound is ..... .



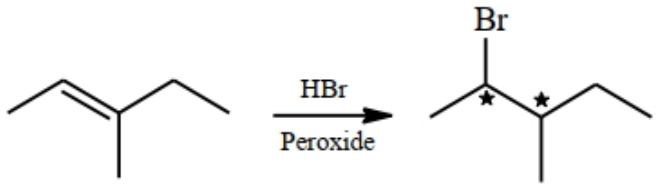
(74) $3-Methylhex-2-$ene on reaction with HBr in presence of peroxide forms an addition product (A). The number of possible stereoisomers for ' A ' is_____.

Solution:

$image$

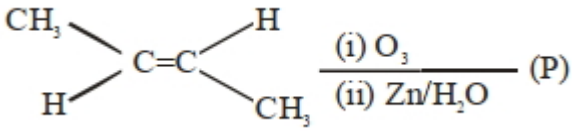
2 chiral centres

No. of stereoisomers $= 4$



(75) $Image$

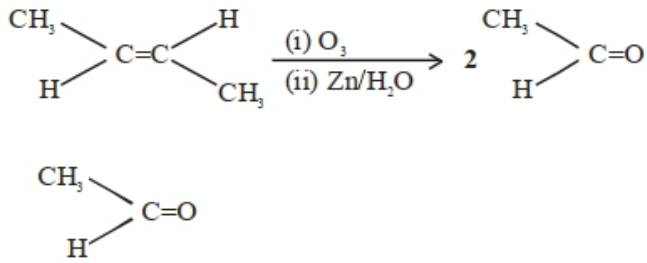
Consider the given reaction. The total number of oxygen atoms present per molecule of the product (P) is_____



Solution:

$Image$

Hence total number of oxygen atom present per Molecule $image$ is I.



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